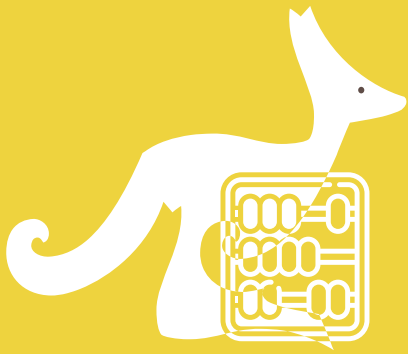




MATH KANGAROO CHINA

考试说明 / Instruction

Time allowed: 75 minutes

考试时长: 75 分钟

Language: English & Chinese

试卷语言: 英文和中文

This is a 30-question multiple-choice competition.

考试包含 30 道单项选择题。

Problems 1 to 10 are worth 3 points each.

1-10 题每题 3 分

Problems 11 to 20 are worth 4 points each.

11-20 题每题 4 分

Problems 21 to 30 are worth 5 points each.

21-30 题每题 5 分

1 point will be deducted for an incorrect answer and

no point for a blank answer.

答错扣 1 分, 不答不扣分。

The full score is 150 (Starting point is 30)

总分为 150 分 (起始分数为 30 分)

Do not open this booklet until you are told to do so.

收到监考老师指令后, 方可打开试卷册。

Mark your answers clearly on the answer sheet using a **2B pencil**.

务必使用 **2B 铅笔**在答题卡上正确填涂自己的答案。

Calculator cannot be used during the competition.

考试期间禁止使用计算器。

DO NOT discuss problems and solutions in any online or public forum until April 30.

4 月 30 日前请勿以任何形式公开讨论试题和答案。

**Grade 9
&
Grade 10**

3 points

1. The leaflet shown includes transparent windows, allowing what is below to be clearly seen when the flaps are folded over.

			4	9	2			
			3	5	7			
			8	1	6			

When both flaps are folded over, what is the sum of the numbers that can be seen through the windows?

如图所示的小册子上带有透明小格，当折叠页面时，可以清楚地看到下面的数字。

请问当左右两侧的页面同时折叠时，通过小格可以看到的数字之和是多少？

- (A) 7 (B) 9 (C) 12 (D) 14 (E) 15

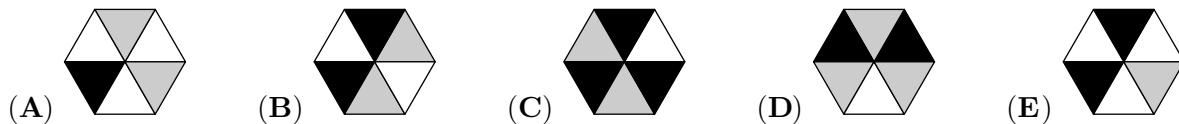
2. The base of a triangle increases by 50% and its height decreases by one-third. What is the ratio of the area of the new triangle to that of the original triangle?

一个三角形的底边长度增加了 50%，同时它的高减少了三分之一。请问这个新三角形的面积与原三角形的面积之比是多少？

- (A) 2:1 (B) 1:1 (C) 1:2 (D) 1:3 (E) 1:4

3. In which of the following hexagons is exactly one-third of the area black and exactly one-sixth of the area white?

请问在以下哪个六边形中，黑色区域的面积恰好占总面积的三分之一，且白色区域的面积恰好占总面积的六分之一？



4. A recipe requires 1 cup of rice and $1\frac{1}{2}$ cups of water. Rouven wants to use $1\frac{1}{2}$ cups of rice. How many cups of water does he need?

一份食谱要求使用 1 杯米和 $1\frac{1}{2}$ 杯水。Rouven 打算使用 $1\frac{1}{2}$ 杯米，请问他需要多少杯水？

- (A) 1 (B) $1\frac{1}{4}$ (C) $1\frac{3}{4}$ (D) $2\frac{1}{4}$ (E) $2\frac{1}{2}$

5. Kangaroo Day takes place every year on the third Thursday of March. Which date is the earliest possible day for Kangaroo Day?

袋鼠活动每年在三月的第三个星期四举行。请问袋鼠活动最早可能在哪一天的哪一天举行？

- (A) 14/3 (B) 15/3 (C) 20/3 (D) 21/3 (E) 22/3

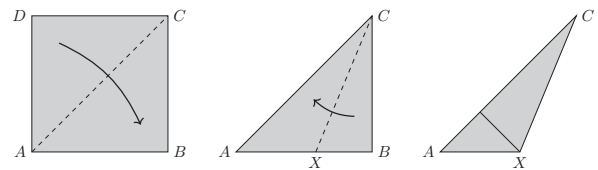
6. Lisa has four wooden digits. She can use them to form the number 2025. How many different numbers greater than 2025 can she form with these digits?

2 0 2 5

Lisa 有四个木制的数字块。她可以利用这些数字块拼出数字 2025。请问她还可以利用这些数字块拼出多少个大于 2025 的不同数字？

- (A) 3 (B) 6 (C) 8 (D) 9 (E) 11

7. Alex folds a square in half along a diagonal to make a triangle. Then he folds the paper again so that one of the short edges of this triangle lies on top of the long edge of this triangle, making the smaller triangle AXC , as shown.



What is the size of angle AXC ?

Alex 先将一张正方形纸沿对角线对折，形成一个三角形。接着，他再次折叠这张纸，使得三角形的一条短边落在其长边上，从而得到了一个更小的三角形 AXC ，如图所示。

请问 $\angle AXC$ 的大小是多少？

- (A) 108° (B) 112.5° (C) 120° (D) 145° (E) 157.5°

8. Luka has some dogs, some rabbits and some cats. Eight of his pets are not dogs. Five of his pets are not rabbits. Seven of his pets are not cats. How many pets does Luka have?

Luka 养了一些宠物，包括小狗、兔子和小猫。在他的宠物中，有八只不是小狗，五只不是兔子，七只不是小猫。请问 Luka 总共养了多少只宠物？

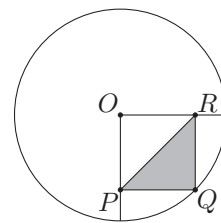
- (A) 10 (B) 11 (C) 15 (D) 16 (E) 20

9. The 4-digit number $80\square\square$ is missing its last two digits. The number is divisible by 8 and 9. What is the sum of these two missing digits?

一个四位数 $80\square\square$ 缺少最后两位数字。已知这个数能同时被 8 和 9 整除，请问这两个缺失的数字之和是多少？

- (A) 6 (B) 8 (C) 10 (D) 12 (E) 14

10. A circle with center O and radius 10 cm is given. A square $OPQR$ is drawn inside the circle, where Q is a point on the circle. What is the area of the shaded triangle PQR ?

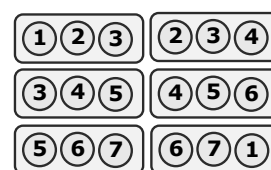


给定一个以 O 为圆心，半径为 10 cm 的圆。在该圆内绘制一个正方形 $OPQR$ ，其中点 Q 位于圆周上。请问阴影三角形 PQR 的面积是多少？

- (A) 12.5 cm^2 (B) 25 cm^2 (C) 50 cm^2 (D) 75 cm^2 (E) 100 cm^2

4 points

11. An athlete has a collection of two gold and five silver medals. They are numbered from 1 to 7, in some order. The picture shows black and white photos of the medals. It is known that in each photo, exactly one of the medals is gold. What is the sum of the numbers on the two gold medals?



一名运动员收藏了两枚金牌和五枚银牌。这些奖牌按照某种顺序从 1 到 7

进行了编号。图中展示的是这些奖牌的黑白照片。已知每张照片中恰好有一枚是金牌，请问这两枚金牌上的编号之和是多少？

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

12. Anna looks at a photo on her smartphone. The format is $16 : 9$ and fills the whole display. When she turns the smartphone, the picture gets smaller. What fraction of the display area is taken up by the smaller picture?



Anna 正在手机上浏览一张照片，照片的比例为 $16 : 9$ ，并且填满整个手机屏幕。当她旋转手机时，照片的显示尺寸会缩小。请问缩小的照片占屏幕面积的比例是多少？

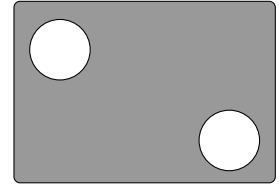
- (A) $\frac{3}{4}$ (B) $\frac{9}{16}$ (C) $\frac{27}{64}$ (D) $\frac{32}{81}$ (E) $\frac{81}{256}$

13. Kate and Tom are celebrating their birthday today. Tom notices that $\frac{1}{19}$ of Kate's age is equal to $\frac{1}{17}$ of his age. The sum of their ages is greater than 40 and less than 100. How old is Tom?

今天是 Kate 和 Tom 的生日。Tom 发现，Kate 年龄的 $\frac{1}{19}$ 恰好等于他年龄的 $\frac{1}{17}$ 。并且，两人的年龄之和大于 40 小于 100。请问 Tom 今年多少岁？

- (A) 19 (B) 34 (C) 38 (D) 57 (E) 76

14. Paul shoots a total of 27 times at two targets. He hits 50% of the shots he aims at the top left target and 80% of the shots he aims at the bottom right target. He misses a total of 9 shots. How many times did he aim for and hit the top left target?



Paul 朝着两个靶子总共射击了 27 次。他朝左上方靶子射击时，命中率为 50%。他朝右下方靶子射击时，命中率为 80%。他总共有 9 次射击未命中。请问 Paul 瞄准并命中左上方靶子的次数是多少？

- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

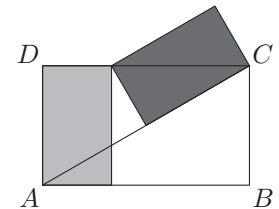
15. Sara has a bag of 18 balls, numbered from 1 to 18. What is the smallest number of balls Sara should remove in order to guarantee that she has removed at least three balls with prime numbers on them?

Sara 的袋子里装有 18 个球，编号为 1 到 18。为了确保取出的球中至少有 3 个球的编号是质数，请问 Sara 至少需要取出多少个球？

- (A) 11 (B) 12 (C) 13 (D) 14 (E) 15

16. The two shaded rectangles are congruent. Both shaded rectangles have area 4. What is the area of the large rectangle?

图中两个阴影矩形是全等的，面积均为 4。请问大矩形的面积是多少？



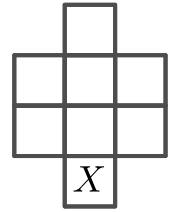
- (A) 10 (B) $8\sqrt{3}$ (C) 8
(D) 12 (E) $4\sqrt{3}$

17. The integer N is the largest six-digit integer with the product of all its digits equal to 180. What is the sum of the digits of N ?

整数 N 是各位数字之积等于 180 的最大六位数。请问整数 N 的各位数字之和是多少？

- (A) 21 (B) 22 (C) 23 (D) 24 (E) 25

18. David wants to place the numbers 1 to 8 in the eight cells of the diagram, with one number in each cell. He wants the cells that contain two consecutive numbers not to share a side or a vertex. Which numbers can David put in cell marked X ?



David 想将数字 1 至 8 填入图中的八个单元格中，每个单元格填一个数字。他希望包含两个连续数字的单元格既不共用一条边，也不共用一个顶点。请问 David 可以在标有 X 的单元格中填入哪些数字？

- (A) 1 or 8 1 或 8 (B) 2 or 7 2 或 7 (C) 3 or 6 3 或 6
(D) 4 or 5 4 或 5 (E) 7 or 8 7 或 8

19. The product of three prime numbers is 11 times their sum. Find the largest possible value that sum could take.

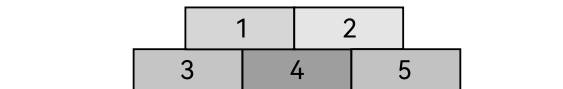
三个质数的乘积是其和的 11 倍。请问这三个质数之和的最大可能值是多少？

- (A) 14 (B) 17 (C) 21 (D) 25 (E) 26

20. Five bricks are placed on the ground, as shown.

Peter can only remove a brick if there are no bricks on

top of it. He selects one of the available bricks at random and removes it, until all the bricks are removed. What is the probability that the brick numbered 4 is the third brick to be removed?

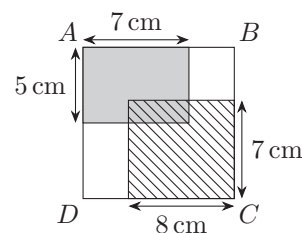


如图所示，地面上放置着五块砖。Peter 只能移除上方没有其他砖块的砖。他每次随机选择一块可以移除的砖块并将其拿走，直到所有砖块都被移除。请问移除的第三块砖是编号为 4 的砖块的概率是多少？

- (A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{1}{5}$ (D) $\frac{1}{6}$ (E) $\frac{1}{8}$

5 points

21. The square $ABCD$ contains two rectangles. One is grey and the other striped, with dimensions as shown in the diagram (not to scale). The area of the overlapping part of the two rectangles is 18 cm^2 . What is the area of $ABCD$?



正方形 $ABCD$ 中包含两个矩形，一个是灰色矩形，另一个是条纹矩形，其尺寸如图所示（未按比例绘制）。两个矩形重叠部分的面积为 18 cm^2 。请问正方形 $ABCD$ 的面积是多少？

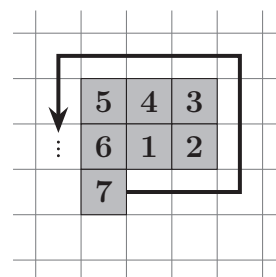
- (A) 64 cm^2 (B) 81 cm^2 (C) 100 cm^2
(D) 121 cm^2 (E) none of the previous 上述答案均不正确

22. $\overline{ABCDEF}$ is a six-digit integer made up of the digits 1, 2, 3, 4, 5, and 6, with no repeated digits. Its first two digits $\overline{AB}$ is multiple of 2, its first three digits $\overline{ABC}$ is multiple of 3, its first four digits $\overline{ABCD}$ is multiple of 4, its first five digits $\overline{ABCDE}$ is multiple of 5, and the full integer $\overline{ABCDEF}$ is a multiple of 6. What is the sixth digit, F ?

$\overline{ABCDEF}$ 是一个六位数，由数字 1、2、3、4、5 和 6 组成，且数位上没有重复数字。它前两位数字组成的数 $\overline{AB}$ 是 2 的倍数，前三位数字组成的数 $\overline{ABC}$ 是 3 的倍数，前四位数字组成的数 $\overline{ABCD}$ 是 4 的倍数，前五位数字组成的数 $\overline{ABCDE}$ 是 5 的倍数，整个六位数 $\overline{ABCDEF}$ 是 6 的倍数。请问第六位数字 F 是多少？

- (A) 2
(B) 4
(C) 6
(D) both 2 and 4 are possible 2 和 4 都有可能
(E) both 4 and 6 are possible 4 和 6 都有可能

23. Daniel numbers certain squares on a sheet of grid paper. Each square has a side-length of 0.5 cm. He starts with one square and then numbers the squares 2, 3, 4, 5, ... in a counter-clockwise direction, as shown. He stops when he has numbered 2025 squares, and looks at the shape made up of all the numbered squares. What is the perimeter of this shape?



Daniel 在一张方格纸上给若干方格编号。每个方格的边长为 0.5 cm。他从一个方格开始编号，然后按逆时针方向依次编号 2、3、4、5、.....，如图所示。当他编号到第 2025 个方格时停止，并观察由所有已编号方格组成的图形。请问这个图形的周长是多少？

- (A) 25 cm (B) 45 cm (C) 80 cm (D) 90 cm (E) 180 cm

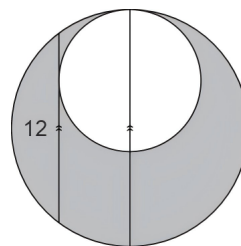
24. A four-digit integer $\overline{ABCD}$ is multiplied by its units digit $\overline{D}$. The result is a different four-digit integer $\overline{DXYA}$, that has the units and thousands digits of the original integer interchanged. How many four-digit integers $\overline{ABCD}$ have this property?

$$\begin{array}{r} A B C D \\ \times \quad D \\ \hline D X Y A \end{array}$$

将一个四位数 $\overline{ABCD}$ 与其个位数字 $\overline{D}$ 相乘，得到一个新的四位数 $\overline{DXYA}$ ，新四位数的千位和个位数字相较于原四位数互换了位置。请问有多少个四位数 $\overline{ABCD}$ 满足这一性质？

- (A) 1 (B) 2 (C) 9 (D) 10 (E) 11

25. In the diagram, the diameter of the inner circle forms part of the diameter of the outer circle. The outer circle has a chord of length 12 that is parallel to its diameter and is also a tangent to the inner circle. What is the area of the shaded region?



如图所示，内圆直径是外圆某条直径的一部分，外圆有一条平行于该直径的弦与内圆相切，长度为 12。请问阴影区域的面积是多少？

- (A) 36π (B) 49π (C) 64π (D) 81π
(E) The information provided is not sufficient. 所提供的信息不足以解答此题。

26. A sequence of numbers $a_1, a_2, a_3, a_4, \dots, a_{10}$ is such that from the third term onwards, each term is equal to the mean of all the previous terms. That is, a_3 is the mean of a_1 and a_2 ; a_4 is the mean of a_1, a_2 , and a_3 ; and so on. In this sequence $a_1 = 8$ and $a_{10} = 23$. What is the value of a_2 ?

一个数列 $a_1, a_2, a_3, a_4, \dots, a_{10}$ 满足从第三项开始，每一项都等于前面所有项的平均值。即 a_3 等于 a_1 和 a_2 的平均值； a_4 等于 a_1, a_2 和 a_3 的平均值，以此类推。已知 $a_1 = 8$ 且 $a_{10} = 23$ ，请问 a_2 的值是多少？

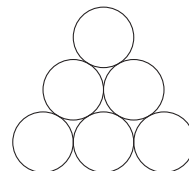
- (A) 28 (B) 32 (C) 38 (D) 44 (E) 50

27. At a party, there are twelve children, including three pairs of twins. How many ways are there to distribute six blue hats and six red hats to the children, so that in each pair of twins, both children are wearing hats of the same colour?

在一个派对上，共有 12 个孩子，其中包括 3 对双胞胎。现在要将 6 顶蓝色帽子和 6 顶红色帽子分发给这些孩子，要求每对双胞胎的两个孩子佩戴相同颜色的帽子。请问有多少种分发方式？

- (A) 72 (B) 86 (C) 92 (D) 102 (E) 132

28. Six circles are arranged in the shape of a triangle, as shown. John writes the digits from 1 to 6 inside the circles so that the sums of the numbers in the circles on all three sides of this triangle are the same. He then calculates the sum of the numbers in the three circles at the vertices of the triangle. How many possible values could he obtain for this sum?

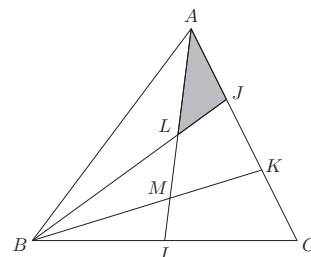


六个圆排列成一个三角形的形状，如图所示。John 将数字 1 至 6 分别填入这些圆中，使得三角形每条边上的圆内数字之和相等。接着，他计算三角形三个顶点处圆内数字的总和。请问该总和可能有多少个不同的值？

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

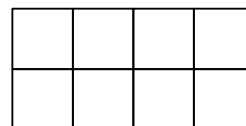
29. Triangle ABC has an area of 60. Point I is the midpoint of side BC , and the points J and K divide side AC into three equal segments. Point L is the intersection of AI and BK . Find the area of triangle ALJ .

已知三角形 ABC 的面积为 60，点 I 为边 BC 的中点，点 J 和 K 将边 AC 分成三等份，点 L 是 AI 和 BK 的交点。请问三角形 ALJ 的面积是多少？



- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

30. Anastasia wants to write the numbers from 1 to 8 into the cells of a 2×4 grid. The number in each cell must be smaller than the number in the cell to its right and smaller than the number in the cell below it. In how many different ways can Anastasia fill the grid?



Anastasia 想要将数字 1 至 8 分别填入 2×4 网格的单元格内。每个单元格内的数字必须小于其右侧相邻单元格的数字，同时也要小于其下方相邻单元格的数字。请问 Anastasia 共有多少种不同的填法？

- (A) 6 (B) 8 (C) 10 (D) 12 (E) 14

Junior

3 points

1. The leaflet shown includes transparent windows, allowing what is below to be clearly seen when the flaps are folded over.

			4	9	2			
			3	5	7			
			8	1	6			

When both flaps are folded over, what is the sum of the numbers that can be seen through the windows?

- (A) 7 (B) 9 (C) 12 (D) 14 (E) 15

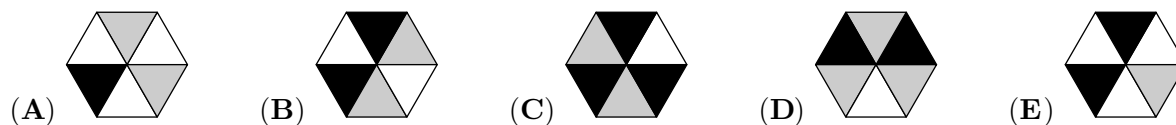
SOLUTION: If you fold the left flap, the left column and the third row are hidden. If you then fold the right flap, the right column is hidden leaving only 9 and 5 visible. Their sum is $9 + 5 = 14$.

2. The base of a triangle increases by 50% and its height decreases by one-third. What is the ratio of the area of the new triangle to that of the original triangle?

- (A) 2:1 (B) 1:1 (C) 1:2 (D) 1:3 (E) 1:4

SOLUTION: Let b be the base of the original triangle and h its height, then its area is equal to $\frac{b \times h}{2}$. The base of the new triangle is $\frac{3}{2} \times b$ and its height is $\frac{2}{3} \times h$. The new area is equal to $\frac{1}{2} \times \frac{3}{2} \times b \times \frac{2}{3} \times h = \frac{b \times h}{2}$ and the ratio is 1.

3. In which of the following hexagons is exactly one-third of the area black and exactly one-sixth of the area white?



SOLUTION: If one third of the hexagon is black that means that 2 triangles are black and if one-sixth of the hexagon is white that means that 1 triangles are white. Picture D is the only one that fullfill these two conditions.

4. A recipe requires 1 cup of rice and $1\frac{1}{2}$ cups of water. Rouven wants to use $1\frac{1}{2}$ cups of rice. How many cups of water does he need?

- (A) 1 (B) $1\frac{1}{4}$ (C) $1\frac{3}{4}$ (D) $2\frac{1}{4}$ (E) $2\frac{1}{2}$

SOLUTION: Rouven needs $1\frac{1}{2} \times 1\frac{1}{2}$ cups of water, i.e. $\frac{3}{2} \times \frac{3}{2} = \frac{9}{4} = 2\frac{1}{4}$.

5. Kangaroo Day takes place every year on the third Thursday of March. Which date is the earliest possible day for Kangaroo Day?

- (A) 14/3 (B) 15/3 (C) 20/3 (D) 21/3 (E) 22/3

SOLUTION: The earliest possible day for the third Thursday of March corresponds with the earliest possible day for the first Thursday which is the 1st of March. When the 1st of March is the 1st Thursday, the 15th of March is the third Thursday.

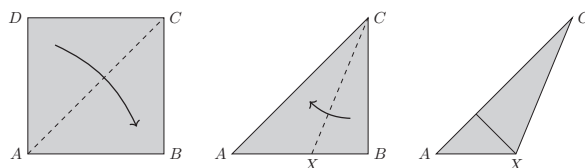
6. Lisa has four wooden digits. She can use them to form the number 2025. How many different numbers greater than 2025 can she form with these digits?

2 0 2 5

- (A) 3 (B) 6 (C) 8 (D) 9 (E) 11

SOLUTION: She can form the numbers 2052, 2205, 2250, 2502, 2520, 5022, 5202, 5220.

7. Alex folds a square in half along a diagonal to make a triangle. Then he folds the paper again so that one of the short edges of this triangle lies on top of the long edge of this triangle, making the smaller triangle AXC , as shown.



What is the size of angle AXC ?

- (A) 108° (B) 112.5° (C) 120° (D) 145° (E) 157.5°

SOLUTION: By construction $\angle CAB = \angle ACB = 45^\circ$. Then $\angle ACX = \frac{1}{2}\angle ACB = 22.5^\circ$. Finally, in $\triangle AXC$, $\angle AXC = 180^\circ - \angle CAX - \angle ACX = 180^\circ - 45^\circ - 22.5^\circ = 112.5^\circ$

8. Luka has some dogs, some rabbits and some cats. Eight of his pets are not dogs. Five of his pets are not rabbits. Seven of his pets are not cats.

How many pets does Luka have?

- (A) 10 (B) 11 (C) 15 (D) 16 (E) 20

SOLUTION:

$$R + C = 8, D + C = 5, D + R = 7$$

$$2(R + C + D) = 20$$

$$R + C + D = 10$$

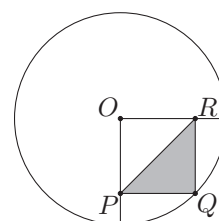
9. The 4-digit number $80\square\square$ is missing its last two digits. The number is divisible by 8 and 9. What is the sum of these two missing digits?

- (A) 6 (B) 8 (C) 10 (D) 12 (E) 14

SOLUTION: Since 8000 is divisible by 8 the two digit must form a number divisible by 8 itself. Because it needs to be divisible by 9, the sum of the digits of the whole number should be divisible by 9. This is true only for 64. So the only 4-digit number divisible by both 8 and 9 starting with 80 is 8064. Hence the sum of the two last digits is 10.

10. A circle with center O and radius 10 cm is given. A square $OPQR$ is drawn inside the circle, where Q is a point on the circle. What is the area of the shaded triangle PQR ?

- (A) 12.5 cm^2 (B) 25 cm^2 (C) 50 cm^2 (D) 75 cm^2 (E) 100 cm^2

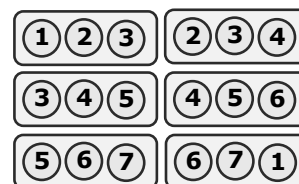


SOLUTION:

Draw segment OQ . We know that $OQ = 10 \text{ cm}$ as it's a radius. Now we get that the area of triangle PQR is equal to half the area of $OPQR$ which is equal to $\frac{10 \times 10}{2} = 50$, hence the area of PQR is 25 cm^2 .

4 points

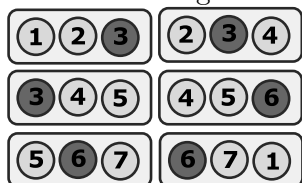
11. An athlete has a collection of two gold and five silver medals. They are numbered from 1 to 7, in some order. The picture shows black and white photos of the medals. It is known that in each photo, exactly one of the medals is gold. What is the sum of the numbers on the two gold medals?



- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

SOLUTION: Solution 1. Since there is one gold in 1,2,3 and one gold in 4,5,6, we conclude that 7 is silver. In a similar way, we conclude that 1 is silver. Since there is one gold in 2,3,4 and one gold in 6,7,1, we conclude that 5 is silver. Then 6 is gold, 4 is silver, 3 is gold and 2 is silver. Finally, the answer is $3+6=9$.

Solution 2. From the first photo we know that one of 1, 2, 3 is gold. Test whether number 1 is gold. If so, from the first photo 2 and 3 are silver. So from the second photo 4 is gold, and then from the third photo it follows that 5 is silver. Next, from the fourth photo 6 is silver and from the fifth 7 is gold. This is impossible because we now have three gold medals, numbers 1, 4 and 7, but we are given there are only two. Similarly, we reject 2 as gold. It remains the 3 is gold, and as above we deduce that the other gold is 6. The figure shows this case.



12. Anna looks at a photo on her smartphone. The format is $16 : 9$ and fills the whole display. When she turns the smartphone, the picture gets smaller. What fraction of the display area is taken up by the smaller picture?

- (A) $\frac{3}{4}$ (B) $\frac{9}{16}$ (C) $\frac{27}{64}$ (D) $\frac{32}{81}$ (E) $\frac{81}{256}$



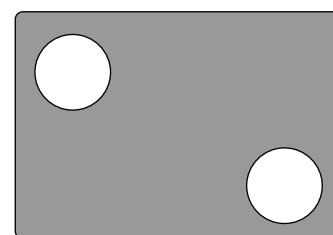
SOLUTION: The width of the photo in the left picture is as long as the long side of the display. And the width of the photo in the right picture is as long as the short side of the display. Therefore, the side lengths of the photo in the right picture are $\frac{9}{16}$ of the side lengths of the photo in the left picture. Hence, the photo on the right takes up $\left(\frac{9}{16}\right)^2 = \frac{81}{256}$ of the whole display.

13. Kate and Tom are celebrating their birthday today. Tom notices that $\frac{1}{19}$ of Kate's age is equal to $\frac{1}{17}$ of his age. The sum of their ages is greater than 40 and less than 100. How old is Tom?

- (A) 19 (B) 34 (C) 38 (D) 57 (E) 76

SOLUTION: Let Kate's and Tom's ages be x and y respectively. Then we have $17 \cdot x = 19 \cdot y$ and hence $x = 19 \cdot k$ and $y = 17 \cdot k$ for some integer k . Using the condition of the problem, we are getting $40 < 17 \cdot k + 19 \cdot k = 36 \cdot k < 100$. It immediately follows, that $k = 2$ is only the integer which satisfies these inequalities, hence Kate is 38 years old and Tom is 34 years old.

14. Paul shoots a total of 27 times at two targets. He hits 50% of the shots he aims at the top left target and 80% of the shots he aims at the bottom right target. He misses a total of 9 shots. How many times did he aim for and hit the top left target?



- (A) 4 (B) 5 (C) 6 (D) 7 (E) 8

SOLUTION: Let a be the number of shots at the top left target, and b the number of shots at the bottom right target. Then we have $a + b = 27$ and $\frac{50}{100}a + \frac{80}{100}b = (27 - 9)$. This leads to $\frac{1}{2}a + \frac{4}{5}(27 - a) = 18$. And therefore $a = 12$. So the answer is 6.

15. Sara has a bag of 18 balls, numbered from 1 to 18. What is the smallest number of balls Sara should remove in order to guarantee that she has removed at least three balls with prime numbers on them?

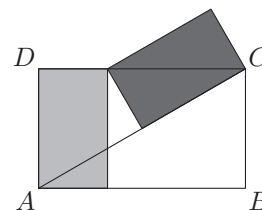
- (A) 11 (B) 12 (C) 13 (D) 14 (E) 15

SOLUTION: There are 7 primes less than 18, that is 2, 3, 5, 7, 11, 13 and 17.

So to guarantee that Sara has removed at least three prime numbers she must have taken all the balls labelled with nonprime numbers, $18 - 7 = 11$ balls and then taken three of the remaining prime numbered balls. This means she has to take $11 + 3 = 14$ balls that is D.

16. The two shaded rectangles are congruent. Both shaded rectangles have area 4. What is the area of the large rectangle?

- (A) 10 (B) $8\sqrt{3}$ (C) 8
 (D) 12 (E) $4\sqrt{3}$



SOLUTION: The lower left gray triangle is congruent to the white triangle in the middle (one leg and 2 angles).

Then the total area above the diagonal of the rectangle we are looking for is the grey rectangle + half of the dark rectangle ($4 + 2 = 6$). It means that the desired area is 12.

17. The integer N is the largest six-digit integer with the product of all its digits equal to 180. What is the sum of the digits of N ?

- (A) 21 (B) 22 (C) 23 (D) 24 (E) 25

SOLUTION: We know that $180 = 2^2 \cdot 3^2 \cdot 5$.

Since N is the largest 6-digit number. and 9 divides 180, the first digit of N is 9.

The product of the last 5 digits of N is $180 : 9 = 20$.

The largest divisor of 20 is 5, so the second digit of N is 5.

The product of the last 4 digits of N is $20 : 5 = 4$.

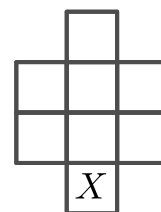
The largest divisor of 4 is 4, so the last 4 digits of N are 4, 1, 1, and 1, respectively. The answer is $9 + 5 + 4 + 1 + 1 + 1 = 21$.

Then the possible value of N is 954111.

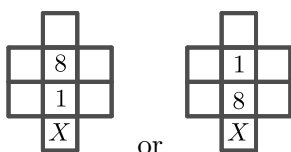
So, the sum of all the constituent numbers of N is $9 + 5 + 4 + 1 + 1 + 1 = 21$.

18. David wants to place the numbers 1 to 8 in the eight cells of the diagram, with one number in each cell. He wants the cells that contain two consecutive numbers not to share a side or a vertex. Which numbers can David put in cell marked X ?

- (A) 1 or 8 (B) 2 or 7 (C) 3 or 6 (D) 4 or 5
 (E) 7 or 8



SOLUTION: The numbers that occupy the central squares must be the ones with the fewest consecutive numbers among the possible, i.e. 1 and 8.



This means that neither the 2 nor the 7 can be in any of the squares of the rows of three squares and so they must go one in the upper box and the other in the lower box.

19. The product of three prime numbers is 11 times their sum. Find the largest possible value that sum could take.

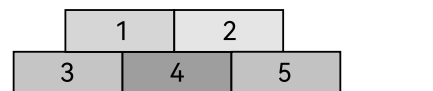
- (A) 14 (B) 17 (C) 21 (D) 25 (E) 26

SOLUTION: Let p , q and r are prime numbers, then we have $pqr = 11(p + q + r)$. Hence one of the prime number has to be 11. Substituting this into the above equation, we are getting $pq = 11 + p + q$. If p and q are both greater than 5, we will get that left hand side is greater than right hand side. So another prime number should be 2 or 3. So we are getting the following triplets satisfying the condition of the problem (2, 11, 13) and (3, 7, 11), hence the maximal value of S is 26.

20. Five bricks are placed on the ground, as shown.

Peter can only remove a brick if there are no bricks on

top of it. He selects one of the available bricks at random and removes it, until all the bricks are removed. What is the probability that the brick numbered 4 is the third brick to be removed?



- (A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{1}{5}$ (D) $\frac{1}{6}$ (E) $\frac{1}{8}$

SOLUTION: To remove the brick N4 we need to remove the first row of bricks (N1 and N2). So we'll need 3 moves.

Case 1. We remove brick N1 (probability $P = \frac{1}{2}$). Then N2 will have to go ($P = \frac{1}{2}$). To have N4 removed we have a chance of $P = \frac{1}{3}$. So the probability for this case is $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{12}$.

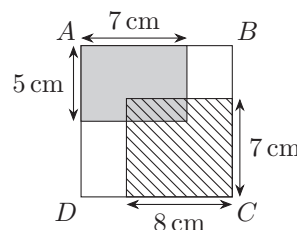
Case 2. Because of the symmetry the same happens if we remove brick N2 first.

Since the events are exclusive the answer is $\frac{1}{12} + \frac{1}{12} = \frac{1}{6}$.

5 points

21. The square $ABCD$ contains two rectangles. One is grey and the other striped, with dimensions as shown in the diagram (not to scale). The area of the overlapping part of the two rectangles is 18 cm^2 . What is the area of $ABCD$?

- (A) 64 cm (B) 81 cm (C) 100 cm
(D) 121 cm (E) none of the previous



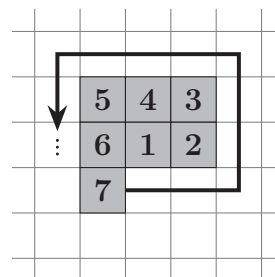
SOLUTION: If the side of the given square is x cm, then the sides of the common part are $(15 - x)$ cm and $(12 - x)$ cm. We have $(15 - x)(12 - x) = 18$ and $x^2 - 27x + 162 = 0$. Only $x = 9$ cm is a solution of the problem and the area of the square is $9 \times 9 = 81$ cm.

22. $\overline{ABCDEF}$ is a six-digit integer made up of the digits 1, 2, 3, 4, 5, and 6, with no repeated digits. Its first two digits $\overline{AB}$ is multiple of 2, its first three digits $\overline{ABC}$ is multiple of 3, its first four digits $\overline{ABCD}$ is multiple of 4, its first five digits $\overline{ABCDE}$ is multiple of 5, and the full integer $\overline{ABCDEF}$ is a multiple of 6. What is the sixth digit, F ?

- (A) 2 (B) 4 (C) 6
 (D) both 2 and 4 are possible (E) both 4 and 6 are possible

SOLUTION: The fifth digit must be 5, the second, the fourth and the sixth digit must be 2, 4 or 6. If the first three digits include 1, 3 and an even number, to be a multiple of 3, the second digit must be 2. The third and fourth digit CD must be multiple of 4, and $C=1$ or 3, therefore D must be 6. Finally, F must be 4

23. Daniel numbers certain squares on a sheet of grid paper. Each square has a side-length of 0.5 cm. He starts with one square and then numbers the squares 2, 3, 4, 5, ... in a counter-clockwise direction, as shown. He stops when he has numbered 2025 squares, and looks at the shape made up of all the numbered squares. What is the perimeter of this shape?



- (A) 25 cm (B) 45 cm (C) 80 cm (D) 90 cm (E) 180 cm

SOLUTION:

Although the figure is drawn in a spiral, at a certain step a square is created, and so we are looking for a square number that is less than or equal to 2025. Noting that $45^2 = 2025$, Daniel stops when his figure is a square with sides of length $45 \cdot 0.5$ cm and circumference $45 \cdot 0.5 \cdot 4 = 90$ cm.

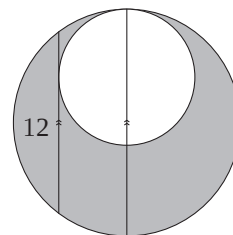
24. A four-digit integer $\overline{ABCD}$ is multiplied by its units digit $\overline{D}$. The result is a different four-digit integer $\overline{DXYA}$, that has the units and thousands digits of the original integer interchanged. How many four-digit integers $\overline{ABCD}$ have this property?

$$\begin{array}{r} A \ B \ C \ D \\ \times \quad \quad D \\ \hline D \ X \ Y \ A \end{array}$$

- (A) 1 (B) 2 (C) 9 (D) 10 (E) 11

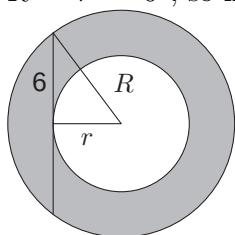
SOLUTION: Since the digit a must correspond to the units-digit of d^2 this in principle marks several possible pairs (a, d) but if we pay attention to the fact that the result must be a four-digits number we see that it must be $d \geq a$ and $d \cdot a < 10$ it turns out that the only possibility is $(a, d) = (1, 9)$. Also it must be $b \cdot d < 10$; so we have $(a, d) = (1, 9)$ and $b = 0$, and then c any value or $(a, d) = (1, 9)$ and $b = 1$ and then c must be 0.

25. In the diagram, the diameter of the inner circle forms part of the diameter of the outer circle. The outer circle has a chord of length 12 that is parallel to its diameter and is also a tangent to the inner circle. What is the area of the shaded region?



- (A) 36π (B) 49π (C) 64π (D) 81π
 (E) The information provided is not sufficient.

SOLUTION: Let the radius of the outer circle be R and the radius of the inner circle be r , so the shaded region has area $\pi(R^2 - r^2)$. Without changing the area, we can move the inner circle to the center of the outer circle, resulting in the scenario as shown in the diagram. By Pythagoras' theorem, $R^2 - r^2 = 6^2$, so hence the shaded region has area 36π .



26. A sequence of numbers $a_1, a_2, a_3, a_4, \dots, a_{10}$ is such that from the third term onwards, each term is equal to the mean of all the previous terms. That is, a_3 is the mean of a_1 and a_2 ; a_4 is the mean of a_1, a_2 , and a_3 ; and so on. In this sequence $a_1 = 8$ and $a_{10} = 23$. What is the value of a_2 ?

- (A) 28 (B) 32 (C) 38 (D) 44 (E) 50

SOLUTION: We have $a_3 = \frac{a_1 + a_2}{2}$ and

$$a_4 = \frac{a_1 + a_2 + a_3}{3} = \frac{a_1 + a_2 + \frac{a_1 + a_2}{2}}{3} = \frac{a_1 + a_2}{2}.$$

Similarly, the subsequent terms of the sequence will be the same. Therefore, $a_{10} = \frac{a_1 + a_2}{2}$, so $23 = \frac{8 + a_2}{2}$. Hence $a_2 = 38$.

27. At a party, there are twelve children, including three pairs of twins. How many ways are there to distribute six blue hats and six red hats to the children, so that in each pair of twins, both children are wearing hats of the same colour?

- (A) 72 (B) 86 (C) 92 (D) 102 (E) 132

SOLUTION: Case 1: all the twins have hats of the same color. This leads to 2 possibilities - all red or all blue.

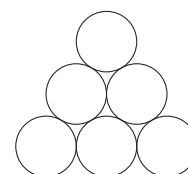
Case 2: exactly two pairs have the same color. This can be done in 3 ways.

We can distribute the other two hats of this color among 6 other children. This can be done in 15 ways: 6 choices for the first child, 5 choices for the second child divided by 2 since the order does not matter. So we have $6 \cdot 5 : 2 = 15$.

This must be multiplied by two because of the symmetry.

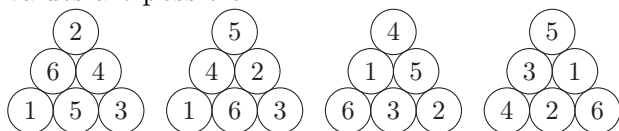
So the total number of ways is $2 + 2 \cdot 3 \cdot 15 = 92$

28. Six circles are arranged in the shape of a triangle, as shown. John writes the digits from 1 to 6 inside the circles so that the sums of the numbers in the circles on all three sides of this triangle are the same. He then calculates the sum of the numbers in the three circles at the vertices of the triangle. How many possible values could he obtain for this sum?



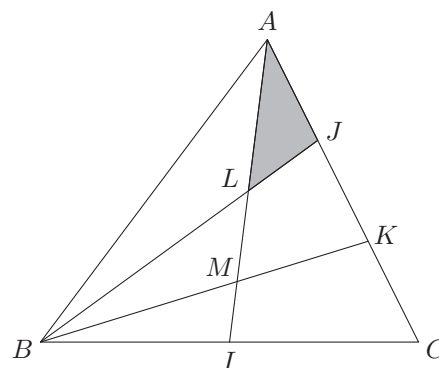
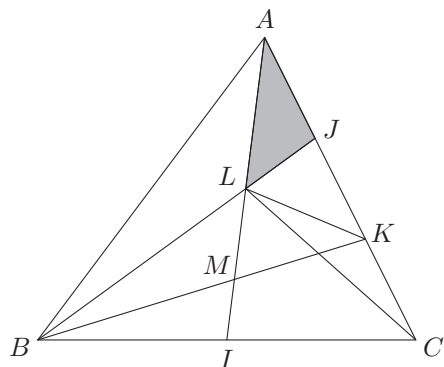
- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

SOLUTION: The sum of the six inscribed numbers is $1 + 2 + 3 + 4 + 5 + 6 = 21$. The sum of the numbers at the vertices of the triangle is not less than $1 + 2 + 3 = 6$, but not more than $4 + 5 + 6 = 15$. The numbers at the vertices are included in the sums of the numbers on the sides of the triangle twice. Therefore, the sum of these three sums is not less than $21 + 6 = 27$, but not more than $21 + 15 = 36$, and is divisible by 3, since this is the sum of three identical numbers. Then the only values that need to be considered for the sums are 6, 9, 12 and 15. The following examples show that all four of these values are possible.



29. Triangle ABC has an area of 60. Point I is the midpoint of side BC , and the points J and K divide side AC into three equal segments. Point L is the intersection of AI and BJ . Find the area of triangle ALJ .

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9



SOLUTION: B I C 1. ****Identifying Equal Areas:**** - The triangles AJL , JLK , and KLC have equal areas, denoted by x . - The triangles LJB and LKI also have equal areas, denoted by y .

2. ****Formulating Equations Based on Area:**** - Since AI is a median, it divides the larger triangles into segments whose areas add up to known values: - The total area of triangle AIC is 30, which can be divided as:

$$3x + y = 30.$$

- The total area of triangle ABC is 60, and triangle BIC (the lower portion) has an area of 40. This portion can be divided as:

$$2x + 2y = 40.$$

3. ****Solving the System of Equations:**** - Using the system of equations:

$$\begin{cases} 3x + y = 30, \\ 2x + 2y = 40, \end{cases}$$

- Solve for x and y : - Simplify the second equation:

$$x + y = 20.$$

- Substitute into the first equation:

$$3x + (20 - x) = 30, 2x = 10, x = 5.$$

- Find y :

$$y = 20 - 5 = 15.$$

****Final Values:****

$$x = 5, \quad y = 15.$$

30. Anastasia wants to write the numbers from 1 to 8 into the cells of a 2×4 grid. The number in each cell must be smaller than the number in the cell to its right and smaller than the number in the cell below it. In how many different ways can Anastasia fill the grid?

- (A) 6 (B) 8 (C) 10 (D) 12 (E) 14

SOLUTION: As no number is larger than 1, the 1 has to be written in the top left cell. Similarly the 8 must be in the bottom right square. For 2 and 7 there are two possibilities, as shown.

1	2		
		7	8

1	2		7
		6	8

1	3		
2		7	8

1	3		7
2		6	8

In the first case in both rows the two missing numbers can be any two of the four remaining numbers in their natural order. There are 6 choices of two out of four numbers, so we have 6 possible fillings in the first case. In the second case, the 6 must be in the bottom row in the second cell from the right. Any of the three remaining numbers fit in the cell above the 6, and the other two fit in the bottom row in their natural order, so we have 3 possible fillings in the second case. The third case is similar to the second case, there are also 3 possible fillings. In the fourth case, the 3 and the 6 must be written in the cells shown. There are two possibilities for the two remaining cells, so we have 2 possible fillings in the fourth case.

In total, there are $6 + 3 + 3 + 2 = 14$ possibilities.